

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 52 | Date: 14-03-2026

Total Questions: 15 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following is typically used to measure the efficiency of an algorithm in terms of its execution time as the input size grows? [2 Marks]

- a) Space Complexity
- b) Time Complexity
- c) Data Complexity
- d) Memory Complexity

2. Which principle does a Stack data structure follow? [2 Marks]

- a) First-In, First-Out (FIFO)
- b) Last-In, First-Out (LIFO)
- c) Last-In, Last-Out (LILO)
- d) First-In, Last-Out (FILO)

3. What does the acronym LIFO stand for, and which common data structure adheres to this principle? [2 Marks]

SECTION A: SHORT ANSWERS

4. What is a Data Structure? [2 Marks]

5. What is the purpose of a pointer in programming? [2 Marks]

6. Give an example of a non-linear data structure. [2 Marks]

SECTION A: MCQS

7. In a singly linked list, which operation has a time complexity of $O(1)$ if the head pointer is known? [3 Marks]

- a) Searching for an element
- b) Deleting the last element
- c) Inserting an element at the beginning
- d) Deleting an element from the middle

8. What is a primary advantage of a linked list over an array for dynamic data storage? [3 Marks]

- a) Direct access to any element by index

- b) Better cache performance due to contiguous memory allocation
- c) Dynamic size, allowing easy insertion and deletion without reallocation overhead
- d) Requires less memory per element

9. The postfix expression for the infix expression $A * (B + C) / D$ is:

[3 Marks]

- a) $ABC+*D/$
- b) $AB*C+D/$
- c) $A+B*C/D$
- d) $A*B+C/D$

SECTION A: SHORT ANSWERS

10. State one significant advantage of a Doubly Linked List over a Singly Linked List.

[3 Marks]

SECTION A: MCQS

11. Explain the Big-O, Big-Omega, and Big-Theta notations used in algorithm analysis.

[5 Marks]

SECTION A: SHORT ANSWERS

12. Differentiate between an array and a linked list based on memory allocation and element access.

[5 Marks]

SECTION A: MCQS

13. What are the primary design considerations when implementing a circular queue using an array? How do these considerations address potential issues?

[6 Marks]

SECTION A: SHORT ANSWERS

14. Outline the steps to insert a new node with data X at the k-th position (1-indexed) in a singly linked list, assuming the head of the list is head.

[6 Marks]

15. Convert the infix expression $(A + B) * C - D / E$ to its equivalent postfix expression, showing the stack and output at each significant step.

[6 Marks]

--- End of Question Paper ---